

ai-os-new

Data Integrity Re-Audit Report

Verification audit after bug fixes. Original: 14 findings. Remaining: 14 findings.

Severity	Count
CRITICAL	2
HIGH	5
MEDIUM	5
LOW	2
Total	14

D-0
1

CRITICAL

MemoryRepository.upsert() always inserts never updates - non-deterministic ID

Files: memory-repository.ts

Problem: computeId() appends crypto.randomUUID(). Every upsert generates new ID. get(id) always returns undefined. Hash uses string lengths not content.

Fix: Use content-based deterministic hash. Remove random UUID suffix.

D-0
2

CRITICAL

IndexedDBStorageDriver silently drops writes when init failed

Files: storage.ts

Problem: If IndexedDB init fails, this.db is null. All set/remove/clear silently return. Caller believes write succeeded. get() returns null indistinguishable from not found.

Fix: Throw error when db is null. Return rejected promise.

D-0
3

HIGH

SettingsService.save() fire-and-forget - data loss on close

Files: settings-service.ts

Problem: save() and saveProfiles() called without await. If user changes settings and closes tab, async setKv may not complete.

Fix: Await save calls. Or register beforeunload handler.

D-0
4

HIGH

PricingService cost history never persisted

Files: pricing-service.ts

Problem: saveHistory() defined but never called. calculateCost() pushes to costHistory but never saves. All cost data lost on close/refresh.

Fix: Call this.saveHistory() after cost history modified. Also save in destroy().

D-0
5

HIGH

Kernel state only saved on beforeunload - lost on crash

Files: kernel.ts

Problem: Periodic auto-save deliberately disabled. State only persisted on beforeunload. If tab crashes or mobile Safari doesn't fire beforeunload, all data lost.

Fix: Re-enable periodic save (30s). Use navigator.sendBeacon or sync localStorage fallback in beforeunload.

D-0
6

HIGH

ConfigHistoryService.history never persisted

Files: config-history.ts

Problem: history array only in memory. No save/persist/database integration. All config history gone on refresh. rollback() useless after refresh.

Fix: Persist history to database.setKv after each commit(). Load in init().

D-0
7

HIGH

SecurityService.changePassword salt update not atomic

Files: security.ts

Problem: After reEncrypt succeeds, new salt written to storageAdapter. If write fails (silently swallowed), masterKey already set to newMasterKey. On restart, old salt + new key = all data unrecoverable.

Fix: Verify salt was written before updating masterKey. Round-trip check.

D-0
8

MEDIUM

Debate session persistence lossy - timestamps/costs zeroed

Files: debate-session-persistence.ts

Problem: sessionToRecord hard-codes totalTokens:0, totalCost:0, startedAt:Date.now(). recordToSession hard-codes maxRounds:10, convergenceScore:0. Round-trip loses all temporal/financial data.

Fix: Preserve original values in sessionToRecord. Restore them in recordToSession.

D-0
9

MEDIUM

ChatStore debounced persistence - no beforeunload guarantee

Files: useChatStore.ts

Problem: Session persistence debounced 1s. visibilitychange:hidden calls flush() but async. No beforeunload. Messages sent <1s before close may be lost.

Fix: Add beforeunload handler with sync localStorage fallback.

D-1
0

MEDIUM

GroupManager.deleteKey partial delete - key orphaned on failure

Files: group-manager.ts

Problem: Persists group state (removing keyId) then calls keyService.removeKey(). If removeKey fails, key removed from groups but still in registry.

Fix: Reverse order: remove from KeyService first, then persist group state.

D-1
1

MEDIUM

importAll uses bulkAdd - crashes on duplicate IDs within import data

Files: dexie-storage.ts, database-service.ts

Problem: All importAll methods use bulkAdd after clearing. bulkAdd rejects duplicate primary keys within import array. Should use bulkPut.

Fix: Replace bulkAdd with bulkPut in all importAll methods.

D-1
2

MEDIUM

MemoryService.prune() cache/DB divergence on individual delete failures

Files: memory-engine.ts

Problem: In importanceBelow pruning, individual delete errors caught with .catch(). In-memory array filtered regardless. On reload, deleted memories reappear.

Fix: Only filter in-memory array for successfully deleted entries.

D-1
3

LOW

Truncated UUID IDs - collision risk at scale

Files: memory-engine.ts, key-registry.ts

Problem: IDs generated as crypto.randomUUID().slice(0,8) - 32 bits. At 10K entries, ~1.2% collision probability. At 100K, ~70%.

Fix: Use at least 12 hex characters (48 bits) or full UUID.

D-1
4

LOW

DebateRepository.clearAll partial clear without transaction

Files: debate-repository.ts

Problem: clearAll() uses Promise.allSettled to clear sessions and verdicts independently. If one succeeds and other fails, inconsistent state.

Fix: Use Dexie transaction for atomicity.